Ej: SI ES POSIBLE, REDEFINIR

$f(x) = \dfrac{16-x^2}{4-x}$ PARA QUE SEA CONTINUA EN $\mathbb{R}$

$f(x) = \dfrac{x^2-16}{x-4}$, $\lim_{x\to 4} f(x) = \lim_{x\to 4} \dfrac{(x+4)\cdot(x-4)}{x-4} = 8.$

REDEFINIMOS

$$\tilde{f}(x) = \begin{cases} \dfrac{x^2-16}{x-4} & \text{si } x \neq 4 \\ 8 & \text{si } x = 4 \end{cases}$$

$$\tilde{f}(x) = x+4$$

Ej: $g(x) = \dfrac{h(x)}{x}$, $\tilde{g}(x) = \begin{cases} \dfrac{h(x)}{x} & \text{si } x \neq 0 \\ 1 & \text{si } x = 0. \end{cases}$

LIMITE ESPECIAL (e):

$$\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x = e \quad , \quad \lim_{y\to 0} (1+y)^{\frac{1}{y}} = e$$

$y = \dfrac{1}{x}$

Ej: $\lim_{x\to+\infty} \left(\dfrac{3x+1}{3x+4}\right)^{\frac{2x^2+1}{x-3}}$ (INDET. 1^∞)

$\dfrac{3x+1}{3x+4} \to 1$, $\dfrac{2x^2+1}{x-3} \to +\infty$

$$= \lim_{x\to+\infty} \left(1 + \frac{3x+1}{3x+4} - 1\right)^{\frac{2x^2+1}{x-3}} =$$

$$= \lim_{x\to+\infty} \left[1 + \frac{3x+1-(3x+4)}{3x+4}\right]^{\frac{2x^2+1}{x-3}} = \lim_{x\to+\infty} \left(1+\frac{-3}{3x+4}\right)^{\frac{2x^2+1}{x-3}} =$$

$$= \lim_{x\to+\infty} \left[\left(1+\frac{1}{\frac{3x+4}{-3}}\right)^{\frac{3x+4}{-3}}\right]^{\frac{-3}{3x+4}\cdot\frac{2x^2+1}{x-3}}$$

$\dfrac{3x+4}{-3} \to +\infty$, $\dfrac{-3}{3x+4}\cdot\dfrac{2x^2+1}{x-3} \to -2$

$\left(1+\dfrac{1}{\frac{3x+4}{-3}}\right)^{\frac{3x+4}{-3}} \to e$

$$= e^{-2}$$

C.A.: $\lim_{x\to+\infty} \frac{-3}{3x+4} \cdot \frac{2x^2+1}{x-3} =$

$= \lim_{x\to+\infty} \frac{x^2}{x^2} \cdot \frac{-3}{3+\frac{4}{x}} \cdot \frac{2+\frac{1}{x^2}}{1-\frac{3}{x}} = -2$

E1: $\lim_{x\to 0} \left(\cos(x)\right)^{\frac{1}{x^2}}$ (INDET. 1^∞) $= \lim_{x\to 0} \left(1+\cos(x)-1\right)^{\frac{1}{x^2}} =$

$= \lim_{x\to 0} \left[\left(1+\cos(x)-1\right)^{\frac{1}{\cos(x)-1}}\right]^{(\cos(x)-1)\cdot\frac{1}{x^2}} = \boxed{e^{-\frac{1}{2}}}$

$(1+y)^{\frac{1}{y}} \xrightarrow[y\to 0]{} e$

C.A.: $\lim_{x\to 0} \frac{\cos(x)-1}{x^2} = \lim_{x\to 0} \frac{(\cos(x)-1)(\cos(x)+1)}{x^2\cdot(\cos(x)+1)} =$

$= \lim_{x\to 0} \frac{\cos^2(x)-1}{x^2\cdot(\cos(x)+1)} = \lim_{x\to 0} \frac{1-\sin^2(x)-1}{x^2(\cos(x)+1)} =$

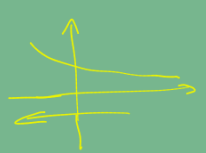
$= \lim_{x\to 0} -\frac{\sin^2(x)}{x^2}\cdot\frac{1}{\cos(x)+1} =$

$= \lim_{x\to 0} -\left(\frac{\sin(x)}{x}\right)^2 \cdot \frac{1}{\cos(x)+1} = -\frac{1}{2}$

E1: $\lim_{x\to 2^-} \left(\frac{2x+2}{5x-2}\right)^{\frac{1}{x-2}} = \boxed{+\infty}$

$\frac{6}{8} = \frac{3}{4} \in (0,1)$

$\left(\frac{3}{4}\right)^x$



Ej: $\lim_{x\to 2^+} \left(\frac{2x+2}{5x-2}\right)^{\frac{1}{x-2}} = 0$

$(x-2 \to 0^+)$ $\quad \frac{2x+2}{5x-2} \to \frac{3}{4}$ $\quad \frac{1}{x-2} \to +\infty$

Ej: $\lim_{x\to+\infty} \left(1+\frac{1}{x}\right)^{x^2} = \lim_{x\to+\infty} \left[\left(1+\frac{1}{x}\right)^{x}\right]^{x} = +\infty$

$\left(1+\frac{1}{x}\right)^{x} \to e$ $\quad x \to +\infty$

Ej: $\lim_{h\to 0} \frac{\ln(2+h)-\ln(2)}{h} = \lim_{h\to 0} \frac{\ln\left(\frac{2+h}{2}\right)}{h} =$

$= \lim_{h\to 0} \frac{1}{h} \cdot \ln\left(\frac{2+h}{2}\right) = \lim_{h\to 0} \frac{1}{h} \ln\left(1+\frac{h}{2}\right) =$

$= \lim_{h\to 0} \ln\left(\left(1+\frac{h}{2}\right)^{\frac{1}{h}}\right) = \lim_{h\to 0} \ln\left(\left[\left(1+\frac{h}{2}\right)^{\frac{2}{h}}\right]^{\frac{1}{2}\cdot\frac{1}{1}}\right) =$

$\frac{2}{h} = \frac{1}{y}$ $\quad y \to 0$ $\quad \left(1+\frac{h}{2}\right)^{\frac{2}{h}} \to e$

$= \ln\left(e^{\frac{1}{2}}\right) = \frac{1}{2}$

Ej: $\lim_{x\to+\infty} \left(\frac{x+2}{x+3}\right)^{x}$ $\quad \frac{x+2}{x+3} \to 1$ $\quad x \to +\infty$

(INDET 1^∞)

$= \lim_{x\to+\infty} \left(1+\frac{x+2}{x+3}-1\right)^{x} =$

$= \lim_{x\to+\infty} \left(1+\frac{x+2-x-3}{x+3}\right)^{x} = \lim_{x\to+\infty} \left(1+\frac{-1}{x+3}\right)^{x} =$

$\frac{-1}{x+3} \cdot x \to -1$ $\quad \frac{-1}{x+3} \to 0$ $\quad \frac{x+3}{-1} = \frac{1}{y}$

$= \lim_{x\to+\infty} \left[\left(1+\frac{-1}{x+3}\right)^{\frac{x+3}{-1}}\right]^{\frac{-1}{x+3}\cdot x}$

$\left(1+\frac{-1}{x+3}\right)^{\frac{x+3}{-1}} \to e$ $\quad y \to 0$

$= e^{-1}$

C.A.: $\lim_{x\to+\infty} \frac{-x}{x+3} = \cdots = -1$

$$\lim_{x\to+\infty}\left(1+\frac{-1}{x+3}\right)^{x} = \lim_{x\to+\infty}\left[\left(1+\frac{1}{-x-3}\right)^{-x-3}\right]^{\frac{1}{-x-3}\cdot x} = e^{-1}$$

($-x-3 \to -\infty$, t; $\frac{1}{-x-3}\cdot x \to -1$; $\left(1+\frac{1}{t}\right)^{t} \to e$)

$$\lim_{t\to\infty}\left(1+\frac{1}{t}\right)^{t}$$

Ej: $\lim_{x\to+\infty}\left(\frac{x+2}{x+3}\right)^{\sqrt[3]{x}}$ (INDET 1^{∞})

$\frac{x+2}{x+3} \to 1$, $\sqrt[3]{x} \to +\infty$

$$= \lim_{x\to+\infty}\left(1+\frac{x+2}{x+3}-1\right)^{\sqrt[3]{x}} = \lim_{x\to+\infty}\left(1+\frac{x+2-x-3}{x+3}\right)^{\sqrt[3]{x}} = \lim_{x\to+\infty}\left(1+\frac{-1}{x+3}\right)^{\sqrt[3]{x}} =$$

($\frac{-1}{x+3} \to 0$)

$$= \lim_{x\to+\infty}\left[\left(1+\frac{-1}{x+3}\right)^{\frac{x+3}{-1}}\right]^{\frac{-1}{x+3}\cdot\sqrt[3]{x}} = e^{0} = \boxed{1}$$

($\frac{x+3}{-1} \to \infty$, $\frac{-1}{x+3}\cdot\sqrt[3]{x} \to 0$; $\left(1+\frac{-1}{x+3}\right)^{\frac{x+3}{-1}} \to e$, $\frac{-1}{x+3} \to 0$)

C. A.: $\lim_{x\to+\infty}\frac{-\sqrt[3]{x}}{x+3} = \lim_{x\to+\infty}\frac{-x^{\frac{1}{3}}}{x^{1}\left(1+\frac{3}{x}\right)} =$

$$= \lim_{x\to+\infty}\frac{-1}{x^{1-\frac{1}{3}}\left(1+\frac{3}{x}\right)} = \lim_{x\to+\infty}\frac{-1}{x^{\frac{2}{3}}\left(1+\frac{3}{x}\right)} = 0$$

($x^{\frac{2}{3}} \to +\infty$, $\left(1+\frac{3}{x}\right) \to 1$)

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